Foundations of Large Language Models

Abilities and Limitations through a Formal Language Learning Perspective

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https://courses.mpi-sws.org/fllm-ws26/

A Shared Inquiry

A language model is designed to *learn* a language

What are the abilities and limitations of large language models (LLMs)?

Theoretical

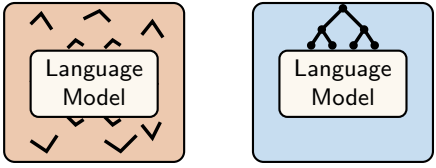
Which languages are *learnable*?

Empirical

Does the prediction *hold* in practice?

Our perspective is through the lens of **formal language** learning

Why Formal Language?



"An ant, viewed as a behaving system, is quite simple. The apparent complexity of its behavior over time is largely a reflection of the complexity of the environment in which it finds itself."

— Herbert Simon, *The Sciences of the Artificial*, 1996

Advantages of Formal Language

A formal language is a set of strings generated by a formal grammar

A probabilistic formal language assigns a non-zero probability to each string in the language

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A probabilistic formal language assigns a non-zero probability to each string in the language

- **Precise specification.** We can decide whether a string belongs to the language
- **Exact sampling.** Train and test strings are sampled i.i.d.
- **No semantic ambiguity, no contamination.** Pure syntax, and unseen in pretraining
- **Control over structure.** The grammar is a knob on complexity

Our Pursuit, in Three Axes

1

No assumption about the computational model

Which languages are learnable at all?

2

A simplified model of the transformer

What can a transformer express and learn?

3

LLMs in practice

How well do LLMs learn a language? What other phenomena do they exhibit?

abstract

concrete

Axis 1: No Assumption about the Computational Model

From positive examples alone, can *any* learner learn a language?

- Can we identify a language in the limit?
- Can we generate new strings from a language in the limit?

Axis 2: A Simplified Model of the Transformer

What is the *computational model* of a transformer?

- What programs can a transformer express?
- Is expressivity a good proxy for learnability?
- If a program is simple, does the transformer learn it?
- Can we derive an interpretable program from the transformer?

Axis 3: LLMs in Practice

How well do LLMs learn a language?
What other phenomena do they exhibit beyond learning?

- What languages are difficult to learn?
- What is the implicit world model of an LLM when learning?
- How do different modes of learning in LLMs compare?
- How does learning relate to memorization?

Logistics

Instructors	Bishwamittra Ghosh, Krishna P. Gummadi
Location	Max Planck Institute for Software Systems, Kaiserslautern, Room 207
Schedule	Weekly, Thursday, 2–4 pm
Audience	PhD and Master's students in computer science
Format	Weekly seminar sessions in which students present and discuss the papers
Prerequisites	No specific coursework required, but a basic understanding of formal languages and LLMs is expected
Paper selection	Submit by 23 October

Inspired by Michael Hahn's prior course, *Theory of Machine Learning* (theory-25)

Grading



- **Presentation.** 45–60 minutes on your paper, then Q&A
- **Report.** At most 2 pages, due before the next class; best three count
- **Participation.** Engagement in the Q&A

Passing

A combined score of **at least 50%**

What We Expect

A good presentation

- Motivate the question and formally state it
- Go deep on one proof or experiment; explain why you find it important
- Explain your takeaways from the paper

A good report on the presented paper

- Discuss at least one aspect of the paper that an AI reviewer, the presenter, or the original authors have missed
- Introduce insights to motivate future research

Class participation

Read the paper **before** the session and ask questions — the more difficult, the better

Syllabus (tentative)

Date	Topic	Reading
Axis 1 — No assumption about the computational model		
TBD	Generation in the Limit	Kleinberg & Mullainathan, *Language Generation in the Limit*, NeurIPS 2024
Axis 2 — A simplified model of the transformer		
TBD	Model of a Transformer	Weiss, Goldberg & Yahav, *Thinking Like Transformers*, ICML 2021
TBD	Predicting Length Generalization	Zhou et al., *What Algorithms Can Transformers Learn? A Study in Length Generalization*, ICLR 2024
TBD	Recovering the Learned Program	Friedman, Wettig & Chen, *Learning Transformer Programs*, NeurIPS 2023
Axis 3 — LLMs in practice		
TBD	Difficulty of Learning	Liu et al., *Exposing Attention Glitches with Flip-Flop Language Modeling*, NeurIPS 2023
TBD	Implicit World Models	Vafa et al., *Evaluating the World Model Implicit in a Generative Model*, NeurIPS 2024
TBD	Learning Modes	Ghosh et al., *Fine-Tuning vs. In-Context Learning in Large Language Models: A Formal Language Learning Perspective*, ACL 2026
TBD	Memorization & Learning	Ghosh et al., *Understanding the Interplay between Memorization and Learning in Large Language Models*, NeurIPS 2026

Next Steps



Before you leave today

Select at least three papers you would be happy to present, by 23 October

Questions?



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